



Chandigarh Engineering College Jhanjeri
Mohali-140307
Department of Artificial Intelligence (AI) and Data Sciences

WashTrack

(Laundry Management System)

BACHELOR OF TECHNOLOGY
(Artificial Intelligence and Data Science.)



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DECLARATION

We, **Ritika** and **Shivangi**, hereby declare that this project report entitled "**WashTrack**" is an authentic record of our own work carried out under the training at **CGC University, Mohali**, in partial fulfillment of the requirements for the award of the **Bachelor of Technology (B.Tech.)** degree in **Computer Science & Engineering**.

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Chapter 1

Introduction

1.1 Project Introduction

The **Laundry Management System** is a web-based application designed to simplify and automate the process of managing laundry services. It enables customers to register, log in, place laundry orders, and track the status of their orders through an easy-to-use interface. The system also provides an admin dashboard to manage customer orders, update order statuses, and monitor overall operations efficiently.

Developed using **Python, Flask, SQLite, HTML, CSS, and JavaScript**, the application reduces manual effort, improves accuracy, and enhances the overall customer experience by providing a secure and organized platform for laundry management.

1.2 Objectives of the Project

- ☐ To automate the laundry order management process.
- ☐ To provide secure user registration and login.
- ☐ To allow customers to place and track laundry orders online.
- ☐ To enable administrators to manage orders through a centralized dashboard.
- ☐ To reduce manual errors and improve operational efficiency.
- ☐ To maintain customer and order information securely using a database.
- ☐ To develop a user-friendly and responsive web application.

1.3 Tools Learned

- ☐ **Python** – Backend programming language.
- ☐ **Flask** – Web framework for application development.
- ☐ **SQLite** – Database for storing user and order information.
- ☐ **HTML** – Structure of web pages.
- ☐ **CSS** – Styling and responsive user interface.
- ☐ **JavaScript** – Client-side interactivity and validation.
- ☐ **Visual Studio Code (VS Code)** – Source code editor used for development.
- ☐ **Git & GitHub** – Version control and project management.



Chapter 2

Software Requirement Analysis

2.1 Problem Definition

Traditional laundry management is primarily carried out using manual methods such as paper records, phone calls, and handwritten receipts. These methods are time-consuming and often result in misplaced orders, inaccurate billing, delayed service updates, and difficulty in maintaining customer records. Customers usually have no convenient way to check the status of their laundry, while administrators face challenges in managing multiple orders efficiently.

2.2 Functional Requirement

- Customer registration and secure login.
- Online placement of laundry orders.
- Real-time tracking of order status.
- Viewing previous order history.
- Admin dashboard for managing customer orders.
- Updating order progress from received to delivered.
- Secure storage and retrieval of customer data.
- Logout functionality for secure session management.

2.3 Non-Functional Requirements

- **Performance:** Processes customer requests quickly with minimal response time.
- **Reliability:** Ensures accurate storage and retrieval of customer and order information.
- **Security:** Uses password hashing and authentication to protect user accounts.
- **Usability:** Provides a clean and simple interface that is easy to understand.
- **Scalability:** Can be extended with additional services, payment gateways, and notification systems in the future.
- **Maintainability:** The modular design allows easy updates and future enhancements.

2.4 Advantages of the Proposed System

The proposed Laundry Management System offers several advantages over traditional manual methods:

- Eliminates manual paperwork and record-keeping.
- Reduces human errors in order management.
- Provides real-time order tracking for customers.
- Improves communication between customers and administrators.
- Ensures secure storage of customer information.
- Saves time by automating routine operations.
- Increases operational efficiency and customer satisfaction.
- Provides a centralized platform for managing all laundry activities.



Chapter 3

Implementation

3.1 System Implementation

The **Laundry Management System** is implemented as a web-based application using the **Flask** framework in Python. The application follows a modular approach, where different functionalities such as user authentication, order management, and database operations are organized into separate components. This makes the application easier to develop, maintain, and enhance.

The frontend provides a simple and responsive interface for customers and administrators, while the backend processes requests, manages data, and communicates with the SQLite database.

3.2 User Registration and Login

The application provides a secure registration and login system for both customers and administrators.

Registration

During registration, users provide their personal details such as:

- Full Name
- Email Address
- Password

The system validates the entered information before creating a new account. Passwords are securely stored in encrypted (hashed) form to improve security.

Login

Registered users can log in using their email address and password. After successful authentication, users are redirected to their respective dashboards based on their role.

3.3 Customer Module

The customer module provides all the essential functionalities required for managing laundry services.

Customers can:

- Place new laundry orders.
- Select laundry service options.
- View previously placed orders.
- Track the current status of ongoing orders.
- Access their account securely.

The module is designed to provide a simple and convenient user experience.

3.4 Admin Module

The administrator has complete control over the management of laundry operations.

The admin dashboard allows administrators to:

- View all customer orders.
- Update the status of laundry orders.
- Monitor completed and pending orders.
- Manage customer information.
- Ensure smooth workflow of daily operations.

This centralized dashboard helps administrators efficiently manage multiple orders simultaneously.



3.5 Database Integration

SQLite is used as the backend database for storing all application data.

The database stores:

- Customer information
- Login credentials
- Laundry order details
- Order history
- Order status

Flask establishes a secure connection with the database to perform Create, Read, Update, and Delete (CRUD) operations whenever required.

3.6 Security Features

Several security measures have been implemented to ensure the safety of user data.

These include:

- Password hashing for secure password storage.
- User authentication before accessing protected pages.
- Role-based authorization for customers and administrators.
- Input validation to prevent invalid or malicious data.
- Secure session management during user login.

These features help maintain the confidentiality and integrity of the application.

3.7 Key Features Implemented

The following features have been successfully implemented in the Laundry Management System:

- User Registration
- Secure Login
- Customer Dashboard
- Laundry Order Placement
- Order Tracking
- Order History
- Admin Dashboard
- Order Status Management
- Database Integration
- Secure Authentication

3.8 System Performance

The developed application performs efficiently by reducing manual effort and automating routine laundry operations. It provides quick access to customer records, minimizes processing time, and allows administrators to manage orders through a centralized interface.

The use of Flask and SQLite ensures fast response time, lightweight deployment, and reliable performance for small to medium-sized laundry business

Chapter 4



Results and Discussion

4.1 Results

The **Laundry Management System** was successfully developed and tested as a web-based application. The system effectively automates the management of laundry services by allowing customers to register, log in, place orders, and monitor their order status through an easy-to-use interface.

The administrator can efficiently manage customer orders, update order statuses, and maintain records through a centralized dashboard. The application stores all information securely in the SQLite database and retrieves it whenever required.

During testing, all major functionalities performed as expected. User authentication, order placement, order tracking, and administrative operations were executed successfully without any significant errors. The integration of Flask with SQLite ensured smooth communication between the application and the database.

The developed system provides a reliable solution for reducing manual paperwork, improving service efficiency, and maintaining accurate customer records.

4.2 Discussion

The implementation of the Laundry Management System demonstrates how web technologies can simplify daily business operations. Compared to traditional manual methods, the developed system offers better organization, improved accuracy, and faster processing of customer requests.

The system provides customers with the convenience of placing orders online and checking their order status at any time. This increases transparency and improves customer satisfaction. At the same time, administrators can efficiently monitor all orders from a single dashboard, reducing workload and minimizing human errors.

The modular design of the application makes it easy to maintain and extend in the future. Additional features such as online payment, delivery tracking, and automated notifications can be integrated without major modifications to the existing system.

The project also highlights the advantages of using Flask and SQLite for developing lightweight web applications. These technologies provide a secure, reliable, and cost-effective platform suitable for small and medium-sized laundry businesses.

4.3 Benefits Achieved

The successful implementation of the Laundry Management System provides several practical benefits:

- Reduced manual paperwork and record maintenance.
- Faster processing of laundry orders.
- Improved accuracy in order management.
- Secure storage of customer information.
- Easy tracking of order status.
- Better communication between customers and administrators.
- Centralized management of laundry operations.
- Enhanced customer satisfaction through timely service updates.



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4.4 Limitations

Although the system performs efficiently, a few limitations still exist:

- The application currently supports only basic laundry management features.
- Online payment functionality has not yet been implemented.
- SMS and email notifications are not available.
- The system is designed for small to medium-sized businesses and may require database upgrades for larger organizations.
- Internet connectivity is required for accessing the web application.

Chapter Summary

The Laundry Management System was successfully implemented and tested, demonstrating its ability to automate laundry service operations efficiently. The system provides secure user authentication, effective order management, and reliable database integration. Overall, the project meets its intended objectives by improving operational efficiency, reducing manual work, and enhancing the customer experience while providing a strong foundation for future enhancements.



Chapter 5

Conclusion & Future Scopes

5.1 Conclusion

The **Laundry Management System** was successfully designed and developed to provide a simple, secure, and efficient solution for managing laundry services. The application automates essential tasks such as customer registration, user authentication, laundry order placement, order tracking, and administrative management through a centralized web-based platform.

The project successfully eliminates many of the challenges associated with traditional manual laundry management, including paper-based record keeping, inaccurate order tracking, and inefficient communication between customers and administrators. By maintaining all customer and order information in a centralized SQLite database, the system ensures data accuracy, reliability, and easy accessibility.

The implementation of Flask as the backend framework and HTML, CSS, JavaScript, and Jinja2 for the frontend resulted in a responsive and user-friendly application. Security features such as password hashing, authentication, and role-based access control further improve the reliability of the system.

Overall, the Laundry Management System achieves its primary objective of simplifying laundry operations, reducing manual effort, improving operational efficiency, and enhancing customer satisfaction. The project also demonstrates the practical application of modern web technologies in solving real-world business problems.

5.2 Future Scope

Although the current system fulfills its intended objectives, several enhancements can be incorporated in future versions to increase its functionality and usability.

The following improvements can be implemented:

- Integration of secure online payment gateways.
- Email and SMS notifications for order confirmation and status updates.
- Mobile application support for Android and iOS devices.
- Home pickup and delivery scheduling features.
- Barcode or QR code-based order identification.
- Customer feedback and rating system.
- Multiple laundry branch management.
- Migration from SQLite to MySQL or PostgreSQL for handling larger databases.
- Advanced reporting and analytics for business performance.
- AI-based demand prediction and service recommendations.

These enhancements will improve the scalability, convenience, and overall performance of the Laundry Management System.



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